Rafael Moretti-Klein

Backend Engineering | Python, FastAPI, Distributed Services

PROFESSIONAL SUMMARY

Backend engineer and former engineering director returning to hands-on individual-contributor work. Experience building high-throughput services in Go and migrating a Python monolith to Python 3 and Go services on GCP. Python remains current through internal tooling and cost-dashboard scripts. Technical strengths include API reliability, Kubernetes, Terraform and observability.

TECHNICAL SKILLS

Languages and frameworks: Python, FastAPI; Go (previous production experience).

Cloud and infrastructure: GCP, GKE, Kubernetes, Terraform; limited AWS experience with EC2 and S3.

Data and service technologies: Cassandra, Redis, BigQuery, gRPC, in-memory caching.

Observability and testing: OpenTelemetry, Prometheus, pytest, tox.

PROFESSIONAL EXPERIENCE

Director of Engineering | Streamhaus GmbH | 2019-August 2026

- Led a platform organisation of 45 engineers.
- Platform work included a strangler-fig migration from a Python 2 monolith to approximately 60 Python 3 and Go services on GKE, spanning 2019-2023.
- Set up a service template using FastAPI or Go, OpenTelemetry tracing, Prometheus metrics and Terraform modules.
- Playback API availability improved from 99.5% to 99.95% during the migration.
- Built an internal cost dashboard using BigQuery billing exports and Looker Studio, with ownership assigned per service. Rightsizing and committed use discounts saved EUR 4.2 million annually (31%) through the 2023-2024 cloud cost programme.
- Recent coding focused on internal tools and Python cost-dashboard scripts.

Principal Engineer | Streamhaus GmbH | 2015-2019

- Technical lead for the playback API; designed and wrote most of the entitlement service determining playback access by subscription, region and device.
- Entitlement service handled 120,000 requests per second at peak with p99 latency below 15 ms, using Cassandra and local in-memory caching.
- Service technologies included Go, gRPC, Cassandra, Redis and Kubernetes.

OPEN SOURCE

Maintainer, Python retry library | 2012-2018

- Maintained a Python retry library with approximately 1,500 GitHub stars; handed over maintenance in 2018.
- Technologies: Python, pytest and tox.